

Directional Gaussian-shell conjectures for weak-event probabilities of causation

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July 18, 2026

Locked parameters. The target is the point-identified monotone probability of necessity or sufficiency. The two directions are PN and PS, with separately indexed factual-probability schedules r_n . Fixed-fold cross-fitting learns the control regression for PN and the treated regression after treatment–outcome swapping for PS.

Abstract

This paper formulates two not-yet-proved directional Gaussian-shell conjectures when the factual probability vanishes but its effective count diverges. The upper conjecture asks whether a cross-fitted score has uniform Gaussian inversion and inverse-square-root effective-count diameter under primitive spline-learning and pointwise score-information conditions. The lower conjecture asks for the same order as an honest-diameter converse on the identical shell. Identification and regular PN/PS variance-inference benchmarks are known support. Finite-sample continuous-law coverage, bounded-count calibration, an all-regime frontier, and post-selection event inference are not asserted.

1 Introduction

Probabilities of causation give a causal interpretation to an observed factual event, but their information can be concentrated in a rare factual stratum. Tian and Wu provide regular semiparametric PN and PS inference, while Zhang, Wang, and Miao study generally partially identified PN bounds under a margin condition. The open object is point-identified monotone inference on a triangular rare-event shell with learned nuisances and an effective-count honest-diameter scale.

Hannart and Naveau construct and select an event from a trajectory by maximizing causal evidence. The PN conjecture instead concerns an

externally fixed event definition and applies only if the resulting continuous-data law belongs to the PN shell; it does not analyze event selection or post-selection validity. The PS conjecture applies separately to a Tian–Wu-style stroke attribution analysis only on its PS shell. The lower conjecture would make directional factual effective count, rather than nominal sample size alone, relevant to Li–Mao–Pearl-style planning.

2 Related work

Tian and Pearl are the foundational identification reference. Tian and Wu’s Lemma 1 on p. 3 gives a modern observed-law PN and PS formulation. For PN, their Section 3.1 and Theorem 1 on p. 4 give the regular efficient influence function and efficiency bound; Section 3.2 on pp. 5–7 gives cross-fitted estimators; and Section 4.2 and Theorem 3 on p. 8 give regular asymptotic normality, efficiency, and plug-in variance estimation. For PS, their Section 7.1, Lemma 2 on p. 15, and Theorem 5 on pp. 15–16 give identification and efficiency bounds, while Section 7.2 and Theorem 6 on pp. 16–17 give asymptotic normality and efficiency. These are known regular benchmarks, not triangular rare-shell diameter results. Cai and Kuroki provide regular variance estimators for PN, PS, and PNS, but not rare-shell studentization.

In the current cited arXiv version, Zhang, Wang, and Miao’s Section 2.1 and Propositions 1–2 on pp. 5–6 derive sharp generally partially identified PN bounds without monotonicity. Their Section 2.2 on pp. 6–7 formulates bound-based confidence intervals, and their Section 3 develops margin-regular inference through Assumption 2 on p. 8, Theorems 1–2 on pp. 8–10, and Proposition 3 with intervals on pp. 11–12. This is inference for nonsmooth sharp bounds, not point-identified monotone PN on a triangular shell, PS, uniform rare-shell score inversion or relative studentization, or an identical-shell causal converse. Chernozhukov et al. supply cross-fitting, Chen supplies sieve tools, and Chernozhukov, Newey, and Singh study debiased global and point-local regression functionals with regularized Riesz representers. Gleser–Hwang’s Theorem 1, in *Annals of Statistics* 15(4):1351–1362, is a general nonexistence result when a limiting law loses an unbounded target. It is only an adjacent weak-information diameter precedent, not a bounded PN or PS, monotone causal, rare-shell-rate, or shell-membership result. Khan–Tamer are an adjacent thin-information reference, and Cuellar–Kennedy target a projected conditional probability rather than this marginal monotone ratio.

3 Formal setup

Let the observed row be independent and identically distributed, with propensity score and treatment-specific outcome regressions. Under consistency, conditional exchangeability, and monotonicity,

$$\Theta_{\text{PN}}(P_n) = 1 - \frac{E[e_n(X)m_{0,n}(X)]}{E[e_n(X)m_{1,n}(X)]}, \quad \Theta_{\text{PS}}(P_n) = 1 - \frac{E[(1 - e_n(X))(1 - m_{1,n}(X))]}{E[(1 - e_n(X))(1 - m_{0,n}(X))]}.$$

PN uses the treated-success factual probability, and PS uses the control-failure factual probability.

4 Assumptions

The row is independent and identically distributed within each triangular-array row. The covariate density and population propensity score are bounded, and the propensity, control regression, and treated regression belong to fixed Hölder balls. Fixed-fold spline learners clip fitted propensities to $[\varepsilon/2, 1 - \varepsilon/2]$ and fitted outcome regressions to $[0, 1]$. Thus, for either direction and $t \in [0, 1]$,

$$|\psi_t^d(O; \hat{\eta})| \leq B_\varepsilon := \frac{2}{\varepsilon} - 1, \quad |\psi_t^d(O; \hat{\eta}) - E\psi_t^d(O; \hat{\eta})| \leq 2B_\varepsilon.$$

An admissible schedule has diverging effective count and the two primitive rare-learning inequalities. Pointwise score information is comparable to the directional factual probability. The construction-specific cap $0 < \bar{d} \leq 1/2$ ensures that the least-favorable type mass $1 - 2r$ is nonnegative; it is not a generic rare-event assumption.

5 Main results

Conjecture 1 (Directional upper frontier (NOT YET PROVED)). *On each direction-specific LF-compatible Gaussian shell, score inversion is uniformly asymptotically Gaussian, has its nominal coverage, and has inverse-square-root effective-count expected diameter.*

Conjecture 2 (Identical-shell lower frontier (NOT YET PROVED)). *Every uniformly honest confidence set over that same directional shell has an inverse-square-root effective-count lower bound on expected diameter.*

The route to these conjectures derives rare-scale nuisance remainders and empirical score-variance consistency. The deterministic envelope gives the Lindeberg control $B_\varepsilon/\sqrt{nr_n} \rightarrow 0$, bounds the centered score in the variance argument, and combines with affine score geometry for the expected-diameter route. The lower route verifies a constant monotone two-point construction in the exact same shell.

Score inversion convention. For direction d , let $\widehat{g}_{d,n}(t)$ and $\widehat{\sigma}_{d,n}^2(t)$ be the cross-fitted score mean and held-out empirical score variance. Define

$$\mathcal{A}_{n,\alpha}^d = \left\{ t \in [0, 1] : n\widehat{g}_{d,n}(t)^2 \leq z_{1-\alpha/2}^2 \widehat{\sigma}_{d,n}^2(t) \right\}, \quad \mathcal{C}_{n,\alpha}^{\text{score},d} = \begin{cases} \mathcal{A}_{n,\alpha}^d, & \mathcal{A}_{n,\alpha}^d \neq \emptyset, \\ \{0\}, & \mathcal{A}_{n,\alpha}^d = \emptyset. \end{cases}$$

This accepts a zero-variance point only when its empirical score is zero. Affinity of the score and continuity of its empirical variance make $\mathcal{A}_{n,\alpha}^d$ closed; the fallback makes the reported set measurable, nonempty, compact-valued, and always of defined diameter. It cannot weaken acceptance of the true value or enlarge diameter.

6 Sanity-check corollaries

When the factual probability is bounded away from zero, the rate conditions reduce to ordinary product-rate conditions and the proposed score formulas agree with Tian–Wu’s PN Sections 3.1–4.2 and Theorems 1 and 3, their PS Sections 7.1–7.2 and Theorems 5–6, and the Cai–Kuroki regular variance-inference context. This is a known sanity reduction, not a new result.

7 Proof-sketch route

The proposed upper proof combines the cross-fit bias identity, monotone localization, the fitted-score envelope, a rare-scale triangular-array central limit theorem, derived studentization, and affine score geometry. The preliminary inversion is closed and the singleton fallback preserves true-value acceptance and cannot enlarge its diameter. The proposed lower proof verifies the capped monotone construction, computes its information ratio, controls two-point product Kullback–Leibler divergence, and applies an honest-confidence-set testing argument.

8 Honest scope flags

The kernel contains only the two conjectures on separate directional Gaussian shells with diverging effective count. Identification and the regular reduction are known. Necessity of rare-learning inequalities and bounded-count diameter are open questions. No finite-sample continuous-law coverage, Poisson limit, all-regime upper frontier, event-selection or selection-honest result, or likelihood-ratio-transfer theorem is asserted.

9 Iteration trail

The current revision adds verified comparator anchors for Tian–Wu, Zhang–Wang–Miao, and Gleser–Hwang, and replaces ambiguous zero-denominator score inversion with a measurable nonempty closed-set inversion and fixed-singleton fallback.

Bibliography

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